

Deeksha Prahlad

+16028166804 [LinkedIn](#) [Work Email](#) [Personal Email](#) [Website](#)

Summary

Ph.D. student in Computer Science at Arizona State University with a research focus on agentic AI for cyber physical systems. My work explores deterministic systems, human-in-the-loop interactions, and the integration of embodied AI in safety-critical applications to ensure reliability and real-time responsiveness.

Education

PhD in Computer Science

Arizona State University, Tempe, AZ

Coursework: Real-time Embedded System, Knowledge Representation

08/2024 - current

Advisor: [Dr. Hokeun Kim](#)

M.S. Robotics and Autonomous systems, Artificial Intelligence

Arizona State University, Tempe, AZ

Coursework: Artificial Intelligence, Artificial Neural Networks, Perception in Robotics, Embedded Machine Learning, Computer Architecture, Statistical Machine Learning, Data Visualization

08/2022 - 05/2024

GPA: 3.77/4

B.E. Electrical and Electronics Engineering

BMS Institute of Technology and Management, Bengaluru, India

Coursework: Control System, Digital Signal Processing, Object Oriented Programming with C++, Python, Power electronics, Digital System Design

08/2017 - 07/2021

GPA: 8.59/10

Skills

Programming Languages: Python, C, R (basic), Git

Coordination Language: Lingua Franca

Cloud Platform: GCP **Tools:** Pytorch, TensorFlow, OpenCV, Keras, Docker, ROS, MATLAB, Tableau, Arduino, IMU sensors, Raspberry Pi

AI and ML: VAE, LSTMs, Computer Vision (CNN, GANs), NLP, SciPy, NumPy, Pandas

Certification: Machine learning, Deep Learning, Generative AI, Reverse and complement nucleic acid sequences using R

Professional Experience

Graduate Research Assistant, Arizona State University, Tempe, Arizona

08/2024- current

Graduate Teaching Assistant, Arizona State University, Tempe, Arizona

08/2024- 12/2025

- Course 420 – Computer Architecture
- Course 230 – Computer Organization and Assembly Programming

Research Intern, Arizona State University, Tempe, Arizona

05/2023- 08/2023

- Implemented reinforcement learning algorithms and physics-informed ordinary differential equation models to predict power grid trajectories with 40% higher accuracy.
- Achieved 98% accuracy rate and significant energy optimization.

Research Intern, Sri Shasha Prayathi Technologies Pvt. Ltd, Bengaluru, India

09/2021- 10/2021

- Achieved a 50% reduction in false positive rates, enhancing system reliability and minimizing unnecessary alerts.
- Demonstrated proficiency in MATLAB for signal visualization and data simulation, showcasing advanced skills in data analysis and representation.

Research Intern, Indian Institute of Science, Bengaluru, India

07/2020- 09/2020

- Literature survey on high voltage techniques in medical applications and conducted an in-depth analysis of data pertaining to 50 diseases and performed statistical data analysis.
- Employed Tableau to assess the effectiveness of virus elimination via electrical voltages, while conducting 10 simulations using MATLAB for further investigation.

Projects

Agentic Driving coach

11/2025- current

- Developed a reactor-MoC using Lingua Franca to enforce determinism in agentic AI-powered CPS.
- The model focused on determinism in the system, consisting of human-in-the-loop, AI agents, and dynamic environments.
- Designed an agentic driving coach as a case study for safety-critical CPS applications.

Personalizing LLM using RAG and KG

07/2024-03/2025

- Developed a RAG framework using knowledge graphs to reduce hallucinations in personalized LLM responses.
- Integrated structured personal data (calendar, contacts, location) for improved contextual understanding.
- Improved response accuracy and personalization compared to baseline LLMs with minimal latency overhead.

Key Point Detection

10/2023-12/2023

- Developed and trained an HRNET model using PyTorch, leveraging the COCO dataset, resulting in a 99% accuracy rate.
- Implemented training of a PyTorch-based HRNET model with a custom dataset, achieving an accuracy rate of 98%.
- Converted the trained model to Tensorflow Lite format and successfully deployed it on a Raspberry Pi running Ubuntu.
- Enabled real-time key point detection feedback, displaying results on the monitor.

Enhancing Large Language Models

09/2023-12/2023

- Optimized LLMs with BERT, actively enhanced the question and answering responses for conversational AI experiences using PyTorch and evaluated it using rouge metrics.
- Improved the efficiency of Llama and T5 models using the transfer learning technique from a pre-trained BERT model for text-to-text generation.

Learning Complex Dexterous Manipulation

01/2023-05/2023

- Employed deep reinforcement learning techniques (DAPG and DAPG++) for tasks like object relocation using Mujoco.
- Implemented behavioral cloning and natural gradient policy techniques, achieving a baseline success rate of 98% using PyTorch.
- Proximal Policy Optimization was added as an improvement to the code to improve the success rate to 99%.

Style GANs Translation

08/2022-12/2022

- Generated authentic images from facial sketches through Image-to-Image translation, providing a solution for limited dataset availability using GANs.
- Leveraged Google Colab GPU resources for model training and inference, significantly reducing training time by 30% using PyTorch.

Awards:

- ACM SIGBED Student Travel Award Recipient for CPS-IoT Week 2026

Publication

- Prahlad, Deeksha, et al. "Agentic Driving Coach: Robustness and Determinism of Agentic AI-Powered Human-in-the-Loop Cyber-Physical Systems", In Proceedings of The 3rd International Workshop on Foundation Models for Cyber-Physical Systems & Internet of Things (FMSys '26), ACM, (To Appear).
- Prahlad, Deeksha, et al. "Personalizing Large Language Models using Retrieval Augmented Generation and Knowledge Graph." Companion Proceedings of the ACM on Web Conference 2025. 2025.
- Lee, Chanhee, et al. "Work-in-Progress: On-device Retrieval Augmented Generation with Knowledge Graphs for Personalized Large Language Models." 2024 International Conference on Embedded Software (EMSOFT). IEEE, 2024.
- Indira Ramarao, Deeksha P, "Applications of Artificial Intelligence with Reference to Hematology: A Review," International Journal of Advances in Engineering and Management (IJAEM), vol. 3, no. 10 Oct 2021, pp. 824–826, Oct. 2021.

References

Dr. Hokeun Kim

Computer Science and Engineering, School of Computing and Augmented Intelligence Arizona State University

Email: hokeun@asu.edu

Website: <https://hokeun.github.io>